Kshitish Ghate

kshitishghate.github.io
kghate@cs.washington.edu

Paul G. Allen School of Computer Science and Engineering
University of Washington
185 E Stevens Way NE, Seattle, WA 98195

♢ RESEARCH INTERESTS

– Training and evaluating LLM agents with realistic, population-grounded simulated users.
– Aligning models to diverse principles and values in human-agent/multi-agent interactions.

♢ EDUCATION

2025 – Present **University of Washington**, Seattle, WA
PH.D. IN COMPUTER SCIENCE & ENGINEERING
Advisors: Aylin Caliskan, Tadayoshi Kohno
Relevant Coursework: Social Reinforcement Learning, Graduate Security and Privacy

2023 – 2025 **Carnegie Mellon University**, Pittsburgh, PA
MASTERS IN LANGUAGE TECHNOLOGIES
Advisor: Mona Diab
Cumulative GPA: 4.0/4.0
Relevant Coursework: Advanced NLP, Introduction to Machine Learning (graduate level), Multimodal ML, On-Device ML, LLM Applications, Question Answering

2018 – 2023 **Birla Institute of Technology and Science, Pilani**, Goa, India
BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND MASTER OF SCIENCE IN ECONOMICS
Cumulative GPA: 9.0/10.0
Relevant Coursework: Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Operating Systems, Foundations of Data Science, Game Theory, Machine Learning, Object Oriented Programming, Probability and Statistics

♢ SKILLS

Languages PYTHON, R, STATA, C/C++, MATLAB, JAVA, SQL
ML/LLM PYTORCH, HF TRANSFORMERS, VLLM, PANDAS, NUMPY, WANDB
Agents & Eval Multi-turn agent and user-simulator pipelines, sandboxed coding-agent evaluation, LLM-as-judge
Infra/HPC Multi-GPU LLM training and large-scale agent rollouts on SLURM HPC clusters; DOCKER, GCP, AWS, AZURE

♢ PUBLICATIONS

— **Preprints & Manuscripts** —

(2026) **Kshitish Ghate**, Michael Saxon, Tadayoshi Kohno, and Aylin Caliskan. "RoleFUZZER: User-Behaviour Fuzzing for Role-Specification Adherence in Large Language Models."

(2026) Harshita Chopra*, **Kshitish Ghate***, Aylin Caliskan, Tadayoshi Kohno, Chirag Shah, and Natasha Jaques. "Beyond Cooperative Simulators: Generating Realistic User Personas for Robust Evaluation of LLM Agents." *Also at the Social Simulation with LLMs Workshop, COLM 2026.*

(in prep.) **Kshitish Ghate**, Chan Young Park, and Jennifer Neville. "DynaPop: Dynamic Population-Grounded User Models for Interactive Agent Evaluation."

— Select Conference and Journal Publications —

2026 Dean Light*, Michael Theologitis*, **Kshitish Ghate***, Shuyue Stella Li, Benjamin Newman, Chirag Shah, Aylin Caliskan, Pang Wei Koh, Dan Suciu, and Yulia Tsvetkov. "Deep Reasoning in General Purpose Agents via Structured Meta-Cognition." *To appear in Advances in Neural Information Processing Systems, 2026. Also at the Workshop on Agent Behavior, COLM 2026.*

2026 **Kshitish Ghate**, Andy Liu, Devansh Jain, Taylor Sorensen, Atoosa Kasirzadeh, Aylin Caliskan, Mona T. Diab, and Maarten Sap. "EVALUESTEER: Measuring Reward Model Steerability Towards Values and Preferences." *Findings of the Association for Computational Linguistics: EMNLP 2026.*

2026 Andy Liu, **Kshitish Ghate**, Mona T. Diab, Daniel Fried, Atoosa Kasirzadeh, and Max Kleiman-Weiner. "Generative Value Conflicts Reveal LLM Priorities." *International Conference on Learning Representations, 2026.*

2025 **Kshitish Ghate**, Tessa Charlesworth, Mona T. Diab, and Aylin Caliskan. "Biases Propagate in Encoder-based Vision-Language Models: A Systematic Analysis From Intrinsic Measures to Zero-shot Retrieval Outcomes." *Findings of the Association for Computational Linguistics, 2025.*

2025 Jingwen Cheng, **Kshitish Ghate**, Wenyue Hua, William Yang Wang, Hong Shen, and Fei Fang. "REALM: A Dataset of Real-World LLM Use Cases." *Findings of the Association for Computational Linguistics, 2025.*

2025 **Kshitish Ghate***, Isaac Slaughter*, Kyra Wilson, Mona Diab, and Aylin Caliskan. "Intrinsic Bias is Predicted by Pretraining Data and Correlates with Downstream Performance in Vision-Language Encoders." *Nations of the Americas Chapter of the Association for Computational Linguistics, 2025.*

2024 Tessa Charlesworth, **Kshitish Ghate**, Aylin Caliskan, and Mahzarin R. Banaji. "Extracting intersectional stereotypes from embeddings: Developing and validating the Flexible Intersectional Stereotype Extraction procedure." *PNAS Nexus, 2024.*

— Select Workshop Publications —

2024 **Kshitish Ghate**, Arjun Choudhry, and Vanya Bannihatti Kumar. "Evaluating Gender Bias in Multilingual Multimodal AI Models: Insights from an Indian Context." *Proceedings of the 5th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*

2024 Nishant Subramani*, **Kshitish Ghate***, and Mona Diab. "Evaluating Personal Information Parroting in Language Models." *4th Workshop on Trustworthy Natural Language Processing.*

* denotes equal contribution

♢ EXPERIENCE

09/25 – Present **University of Washington**, Seattle, WA
Graduate Research Assistant, Paul G. Allen School of Computer Science and Engineering
Advisors: Aylin Caliskan, Tadayoshi Kohno

– Built RoleFUZZER, a coverage-guided user-behaviour fuzzing framework testing whether role-conditioned LLMs adhere to their role specifications in multi-turn interactions, reaching $\geq$89% behaviour-grid coverage vs. 54% for unguided LLM-generated users.
– Developed Persona Policies, an evolutionary program search for realistic user-simulator personas. Simulated users were judged human 80% of the time (2$\times$ baseline), and agents fine-tuned on them improved task success with real users by 23%.
– Co-led DOLORES, a general-purpose agent that builds just-in-time harnesses via structured meta-reasoning, outperforming the strongest harness baseline by 24.8% on average.

06/26 – 09/26 **Microsoft Research**, Redmond, WA
Research Intern, AI Interaction and Learning
Mentors: Chan Young Park, Jennifer Neville

– Built DynaPop, which induces an interpretable user-behaviour space from real user-AI logs (WildChat, SWE-Chat) and fits a population-grounded transition model to simulate synthetic users grounded and consistent with the population they are sampled from.
– Evaluated coding agents on sandboxed software-engineering tasks against these users, producing a vulnerability map that attributes agent failures (e.g., premature completion claims, ignored corrections) to diverse user behaviour types.

09/23 – 08/25 **Carnegie Mellon University**, Pittsburgh, PA
Graduate Research Assistant, Language Technologies Institute
Advisors: Mona Diab, Maarten Sap

– Developed EVALUESTEER, a controlled synthetic benchmark evaluating LLM and reward-model (RM) steerability to user values and styles, revealing a >25-point steerability gap in RMs.
– Created CONFLICTSCOPE, a fully-automated evaluation pipeline generating 1K+ value-conflict scenarios, and improving alignment consistency under conflict by 14% through prompt steering.
– Investigated the relationship between intrinsic biases in 131 CLIP models, their pretraining factors and demonstrated the propagation of representation biases to downstream retrieval tasks.
– Devised a harm reduction framework for reducing hallucinations and improving accuracy in LLM responses used for clinical decision-making through counterfactual synthetic data generation.
– Introduced a risk taxonomy for Personal Information memorization in LLMs and improved existing detectors with 90% better FPR.

08/22 – 12/22 **Amazon**, Bangalore, India
Applied Scientist Intern, Alexa – AI Natural Language Understanding
Supervisors: Anurag Dwarakanath, Anjali Shenoy

– Implemented a novel training methodology and model architecture, drawing from Curriculum Learning literature, to address problem of classifying long tail data in NLU tasks.
– Achieved 5% improvement in F1 score and Intent Classification accuracy by applying a holistic sample difficulty metric in training.

♢ TEACHING

Fall 2024 Large Language Models: Methods and Applications (11-667), Teaching Assistant, CMU
Spring 2022 Applied Econometrics, Teaching Assistant, BITS Pilani

♢ ACADEMIC SERVICE

Organizing Co-organizer, Pluralistic Alignment Workshop, ICML 2026
Reviewing ACL Rolling Review, ICLR, NeurIPS

♢ HONORS & AWARDS

2026 Travel Funding awarded by Simile, Social Simulation with LLMs Workshop, COLM 2026
2018 - 2023 National Talent Search Examination (NTSE) Scholarship, NCERT